Development of Pharyngeal Apparatus..

Learning Objectives.

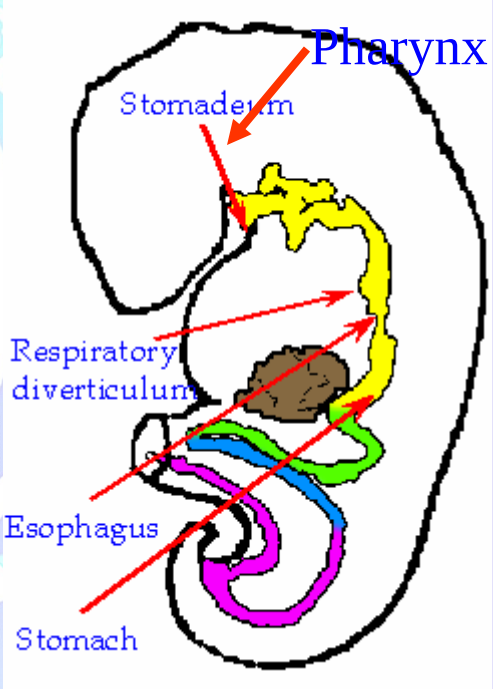
- At the end of lecture, students should be able to know:
- Definition of pharyngeal Apparatus
- Describe Development of pharyngeal Apparatus
- Name the different Parts of pharyngeal Apparatus
- Discuss Fate of pharyngeal Arch
- Discuss Fate of pharyngeal Cleft
- Discuss Fate of pharyngeal Pouch
- Discuss Fate of pharyngeal Membrane
- Describe the Congenital Anomalies related to pharyngeal Apparatus.

PHARYNGEAL APPARATUS.

- **PHARYNGEAL APPARATUS** is the structure which develops in embryo.
- It is comparable to gills of fish.
- But in human the Branchial Apparatus developed into other structures different from gills.
- First indication for development of Branchial Apparatus is, at 4th week the Neural Crest cells invade future head and neck region of embryo.
- cells form ridges on side of head and neck. Located lateral to rostral part of the foregut; will form branchial Apparatus.

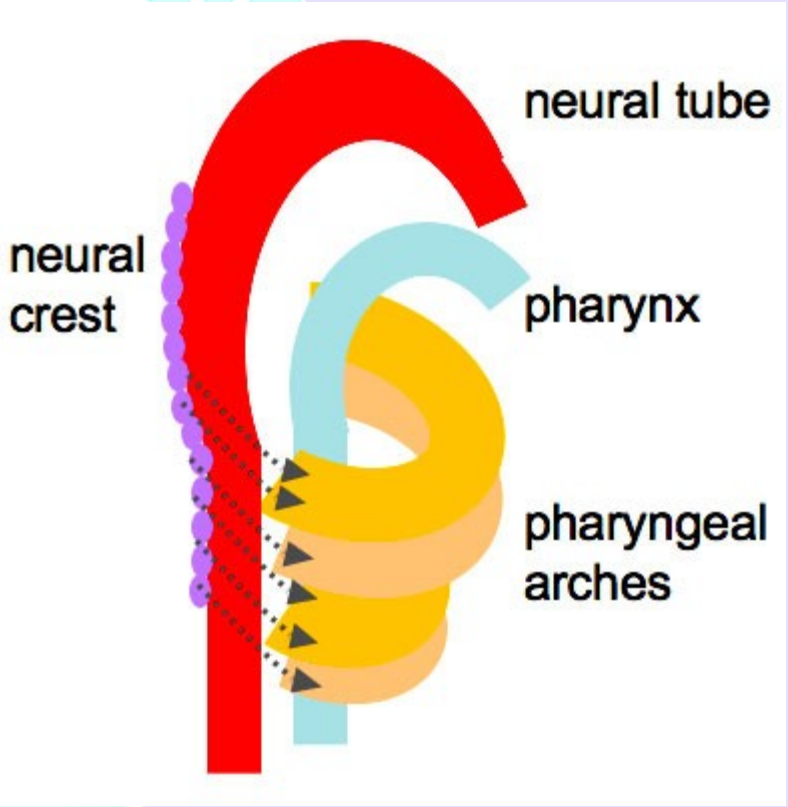
PHARYNGEAL APPARATUS.

- During 5th week the Stomadeum is Present, which lined with Ectoderm and it develops by the proliferation of Mesenchyme.
- The oropharyngeal Membrane initially separates the Stomadeum and Foregut (Pharynx).
- Later it will degenerate allowing communication between Stomadeum and Foregut.



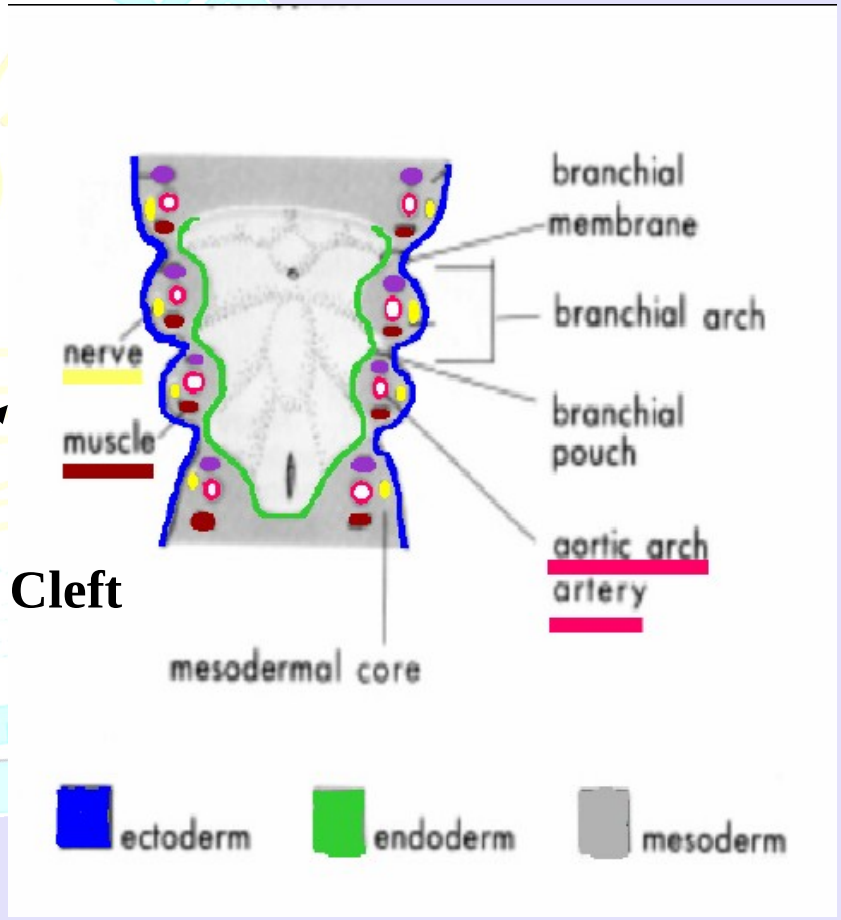
Appearance of Pharyngeal Arches.

- During the fourth week of embryonic development, a series of 5 bar-like ridges appear n the lateral walls of the anterior part of the fore-gut (Pharynx).
- These are called as Branchial or Pharyngeal Arches.
- These Bar like ridges are formed by the migration of Neural Crest Cells into Cervical Region



Components of Branchial Apparatus

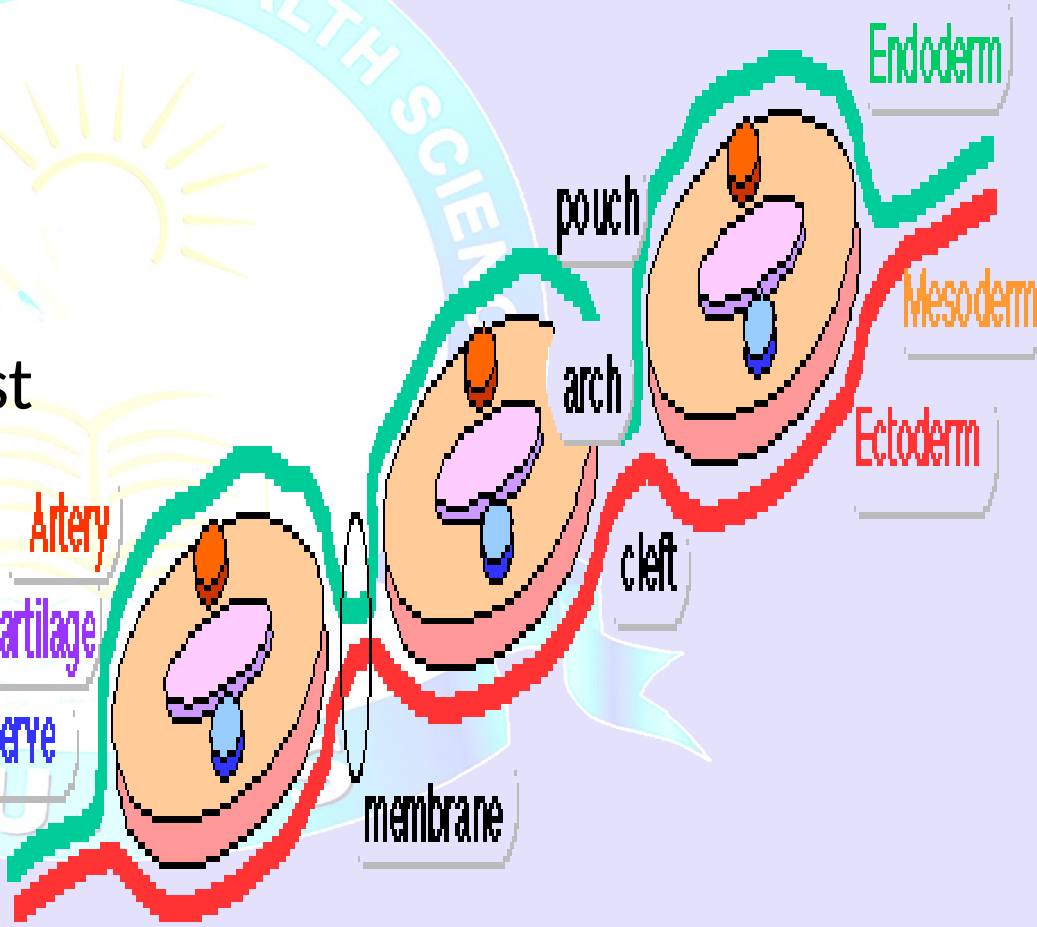
- Brachial Arches,which develop in series one above another in the lateral wall of Primitive Pharynx.
- **Each Consists of Four Components**:
- Pharyngeal Arch
- Pharyngeal Pouch
- Pharyngeal Groove
- Pharyngeal Membrane



Branchial Cleft

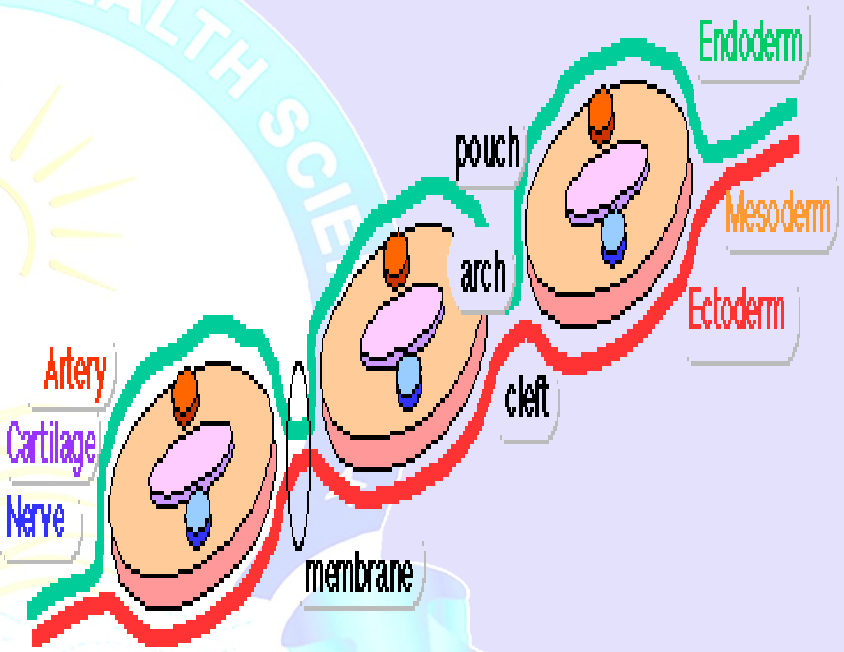
Pharyngeal Arch

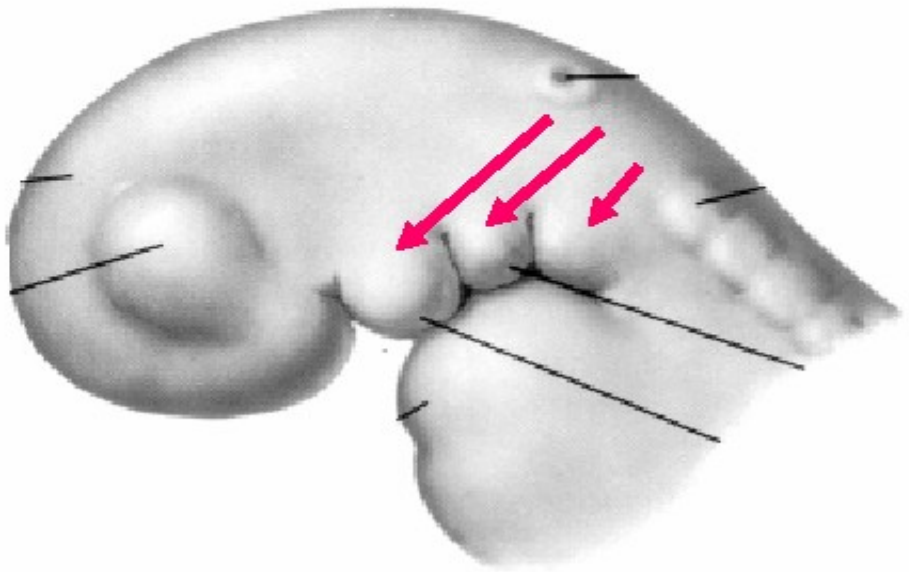
- Each arch has initially similar components
- contributions from all 3 germ layers
  - Ectoderm - outside surface and Neural Crest of core
  - Mesoderm - core of mesenchyme
  - Endoderm - inside pharyngeal surface



Pharyngeal Arch Features

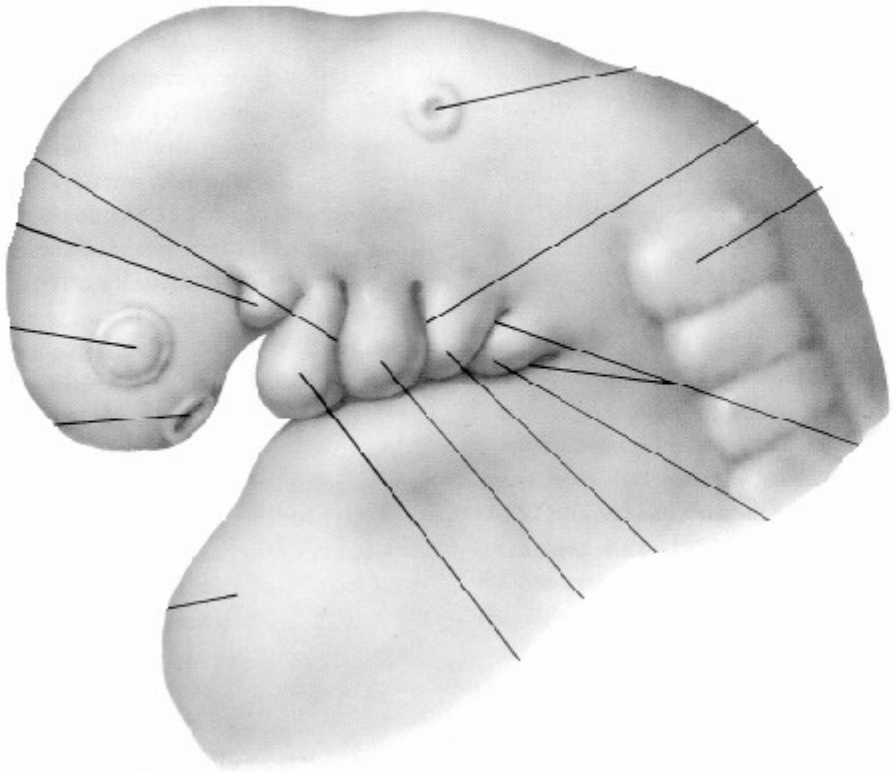
- Arch
- Groove/cleft - externally separates each arch
- Pouch - internally separates each arch
  - Pockets from the pharynx
- Membrane - ectoderm and endoderm contact regions





Neural Crest Cells Invade Head and Neck Lateral To Rostral Part of Foregut

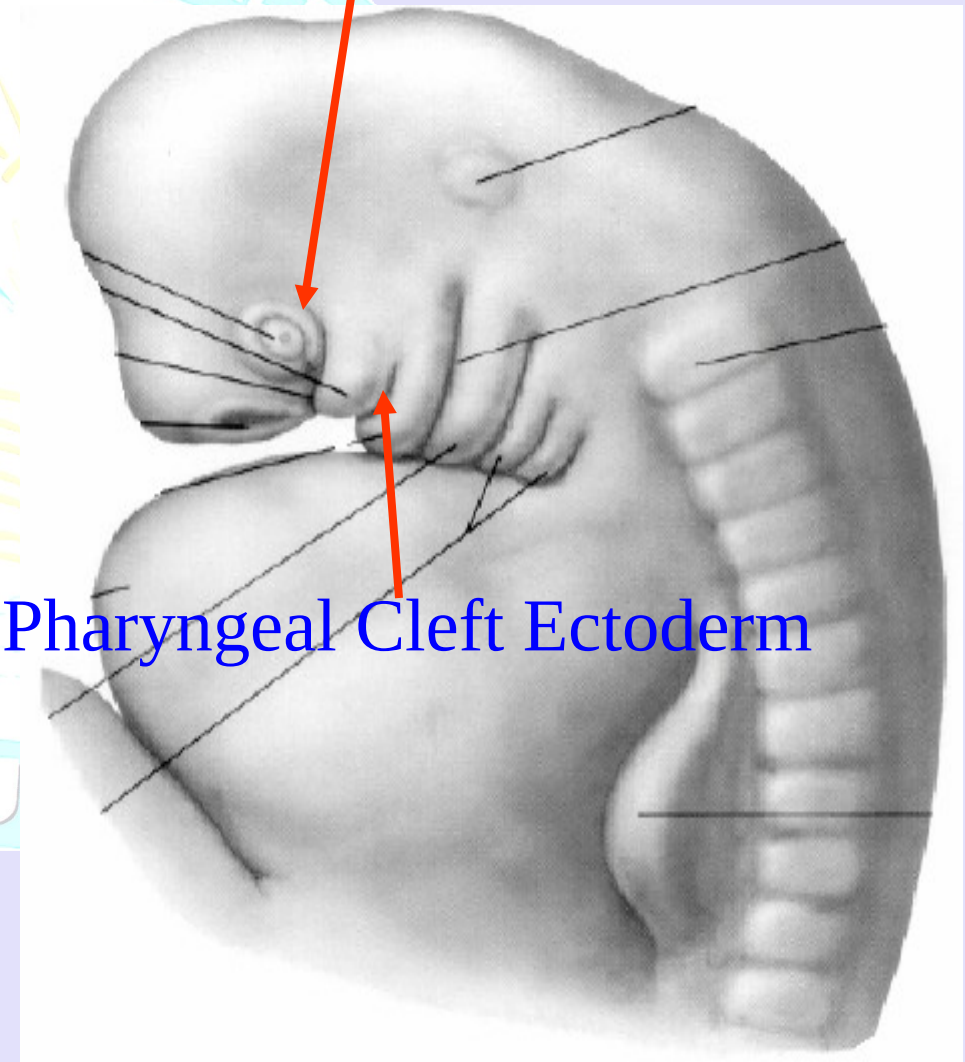
Ridges = Branchial Arches



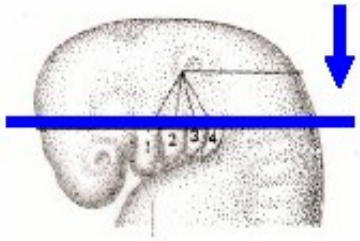
Branchia
Means Gill
In Greek;
In fish, similar structures form Gills

Branchial Apparatus

- Structures in Embryonic Branchial Arches reorganize to form:
- cartilages, nerve, muscles
- & arteries in fetus.
- Forms much of musculature of head and some of neck at 5- 6 weeks

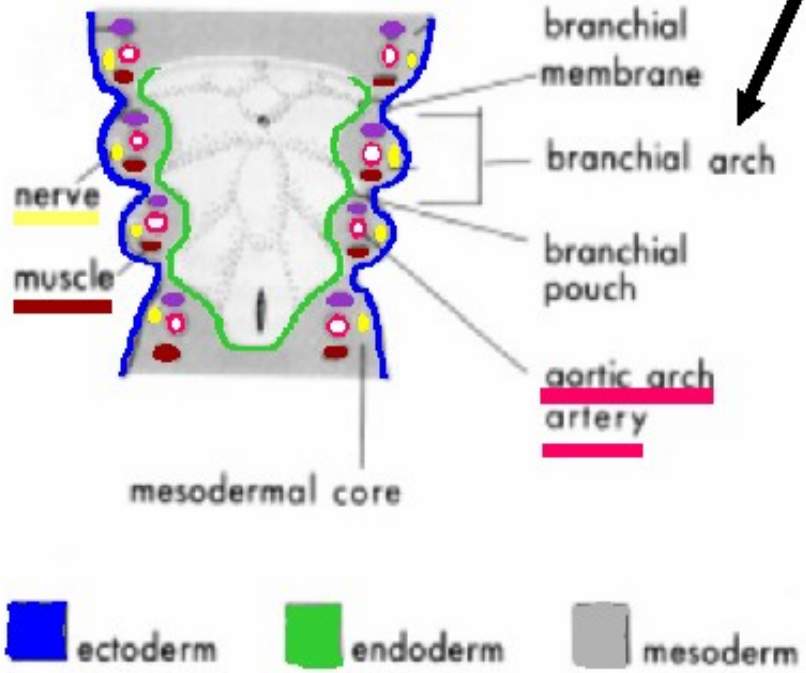


B. BRANCHIAL APPARATUS - 4 elements



ORIENT: LOOKING DOWN

PLANE OF CUT

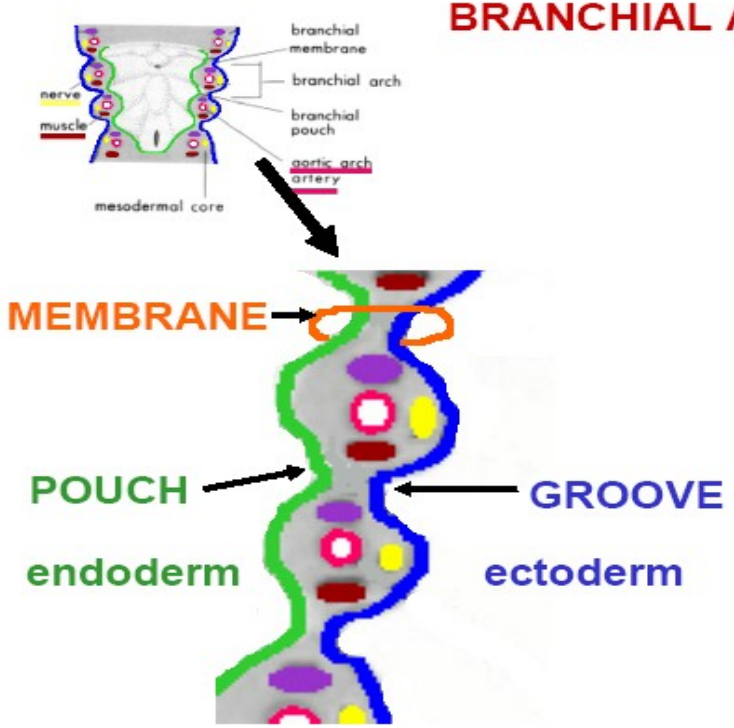


1. **Branchial Arch**
covered by:
Ectoderm - externally
Endoderm - lined internally
(Mesenchyme-core)

Each arch has own cartilage, nerve, muscle and artery (= aortic arch artery)

Each nerve innervates structures derived from its associated arch

BRANCHIAL APPARATUS - 4 elements

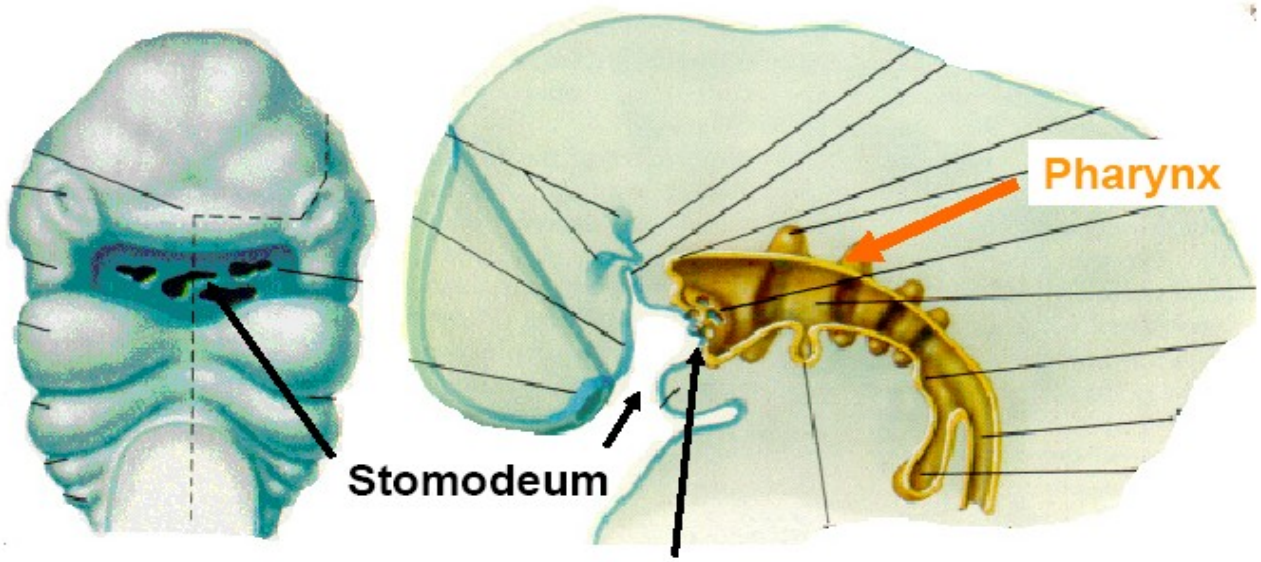


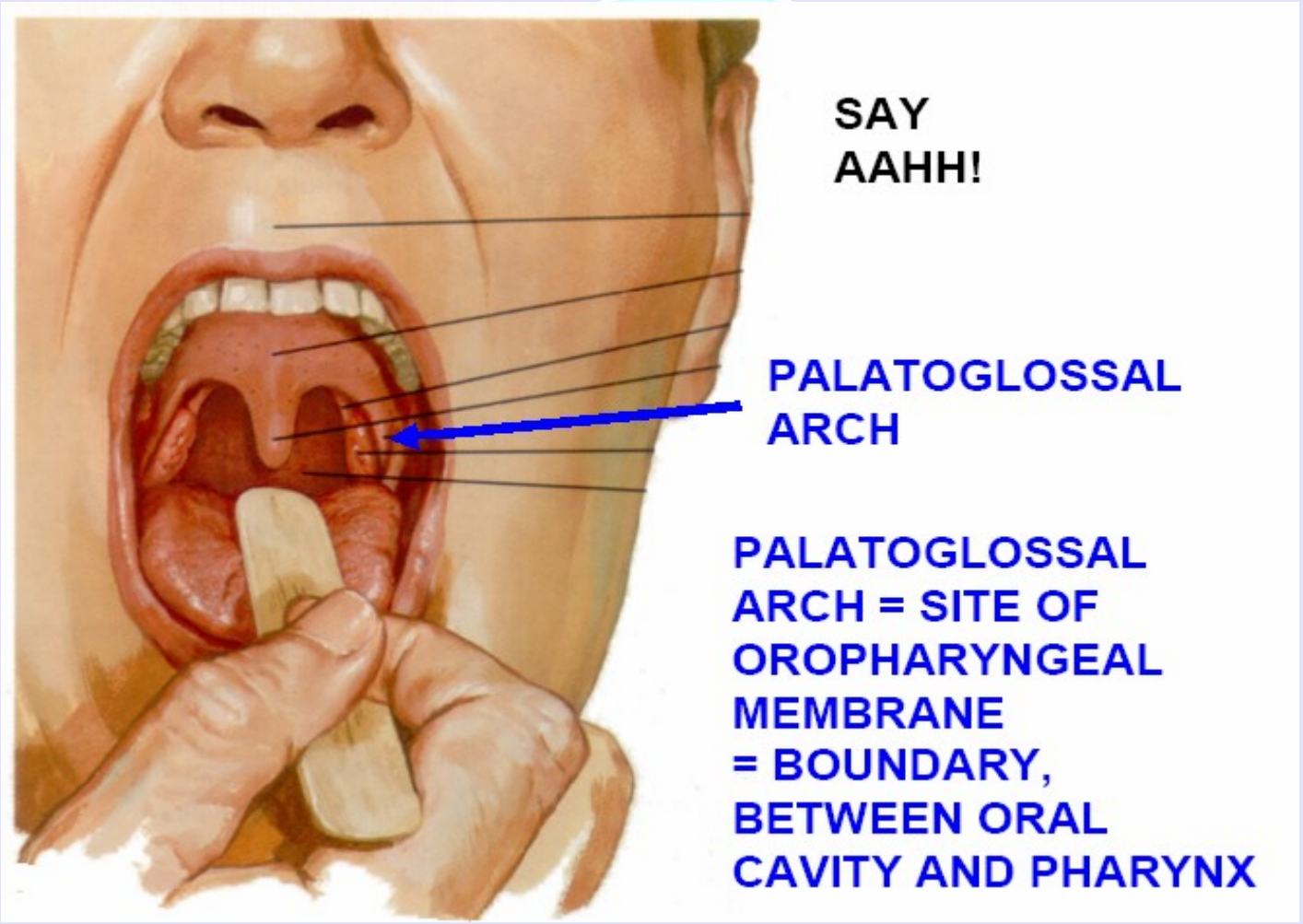
2. Branchial Groove (Pharyngeal Cleft)
- ectodermal cleft between adjacent arches

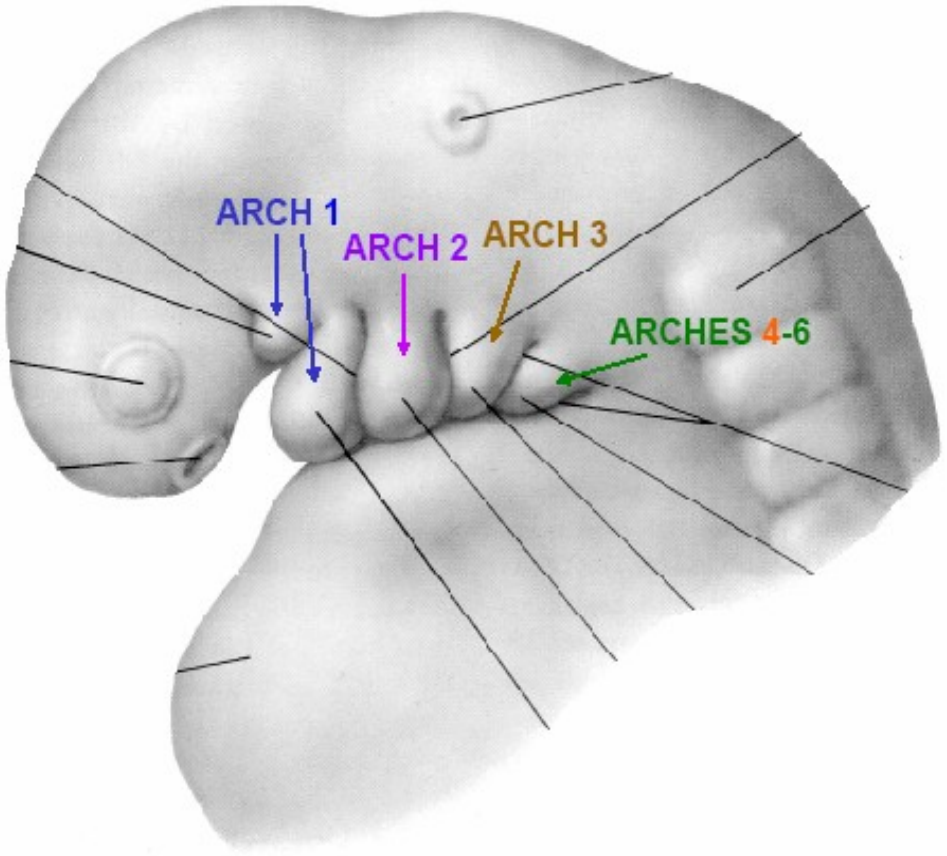
3. Branchial Pouch - endodermal outpocketing from rostral foregut
-between adjacent arches

4. Branchial Membrane
- site of contact of
Groove (ectoderm)
Pouch (endoderm)

- Stomodeum formed by Ectoderm; forms Oral Cavity & Nasal Cavity
- Contacts Endoderm at Oropharyngeal Membrane
- Pharynx – rostral foregut - formed by Endoderm




SAY
AAHH!
PALATOGLOSSAL
ARCH
PALATOGLOSSAL
ARCH = SITE OF
OROPHARYNGEAL
MEMBRANE
= BOUNDARY,
BETWEEN ORAL
CAVITY AND PHARYNX

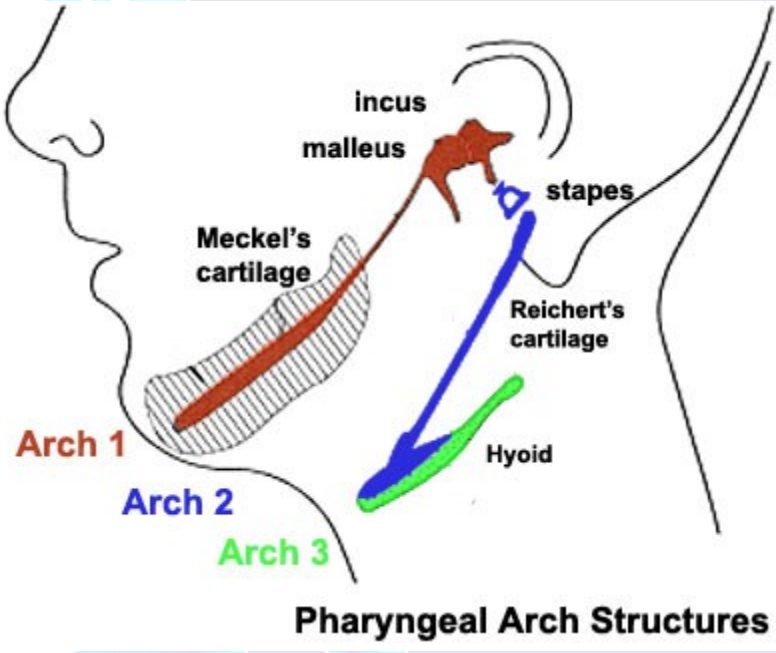


Note:
All authors agree on:
1) Fate of Arches 1- 3;
2) Arch 5 does not form structures in humans

Accounts vary on Arches 4 and 6 (6 is small)

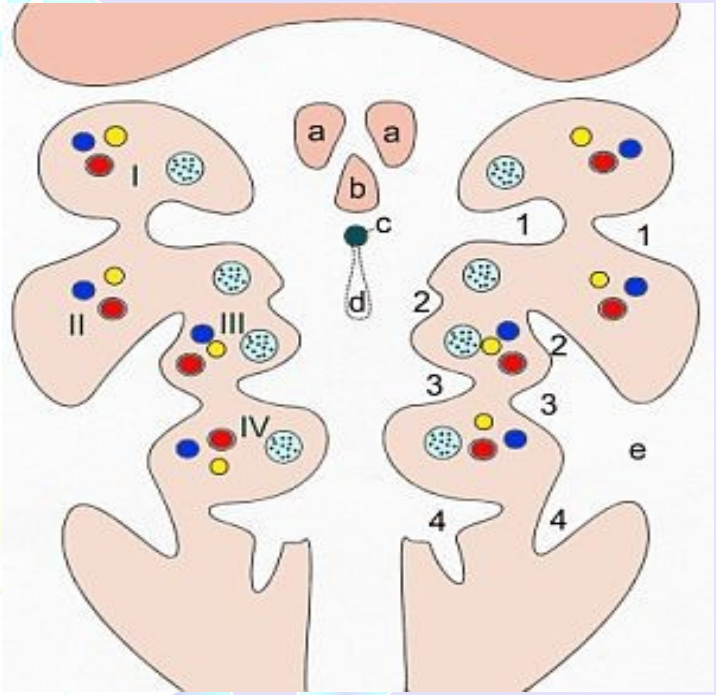
Fate of Pharyngeal Arches

- There are six pharyngeal arches, but in humans the fifth arch only exists transiently during embryologic growth and development.
- Since no human structures result from the fifth arch, the arches in humans are I, II, III, IV, and VI.

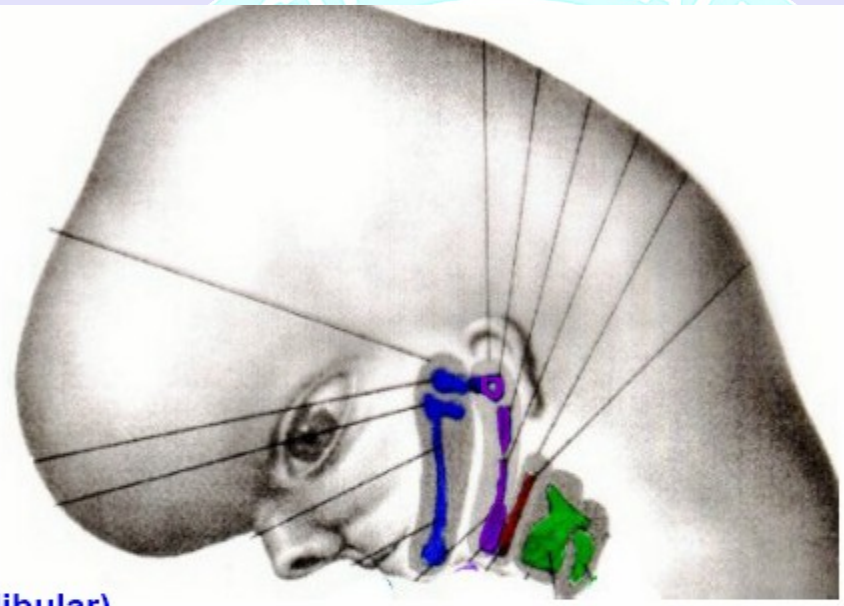


Fate of Pharyngeal Arches

- The structures are developed from all three basic germ layers.
- Each Branchial Arch has its own nerve, these are cranial nerves.
- First arch = Trigeminal (V).
- Second arch = Facial N. (VII).
- Third arch =Glossopharyngeal N. (IX).
- Fourth arch = Vagus (X).
- Sixth arch = Accessory N. (XI).



PHARNGEAL ARCH CARTILAGES.



I First (Mandibular) Arch -
1. Malleus
2. Incus
3. Ant. Ligament Of malleus
4. Sphenomandibular ligament

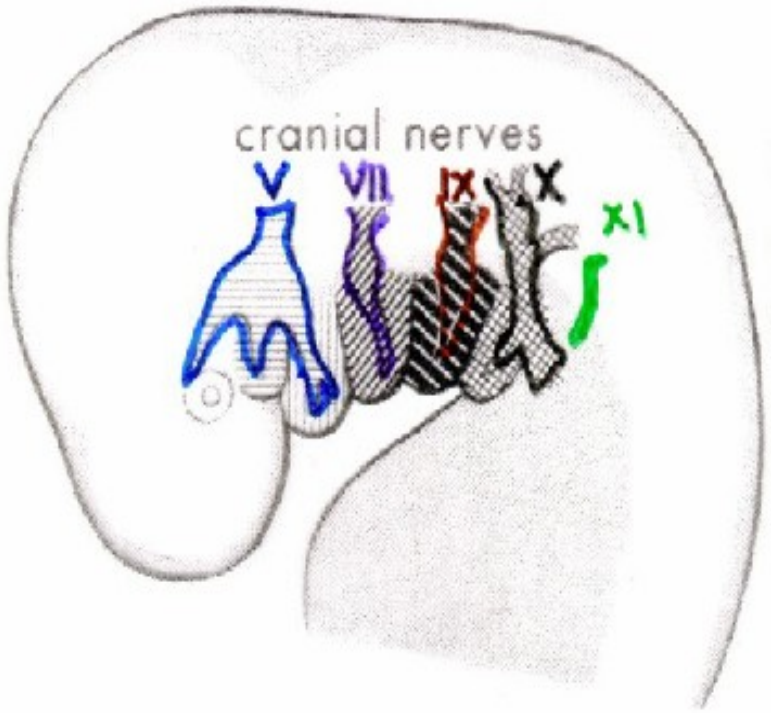
II Second (Hyoid) Arch
1. Stapes
2. Styloid Process
3. Stylohyoid Ligament
4. Lesser horn, Upper ½ body Hyoid

III Third Arch - Lower ½ Body, Greater Horn Of hyoid

IV Fourth (Sixth) Arch - Cartilages Of larynx

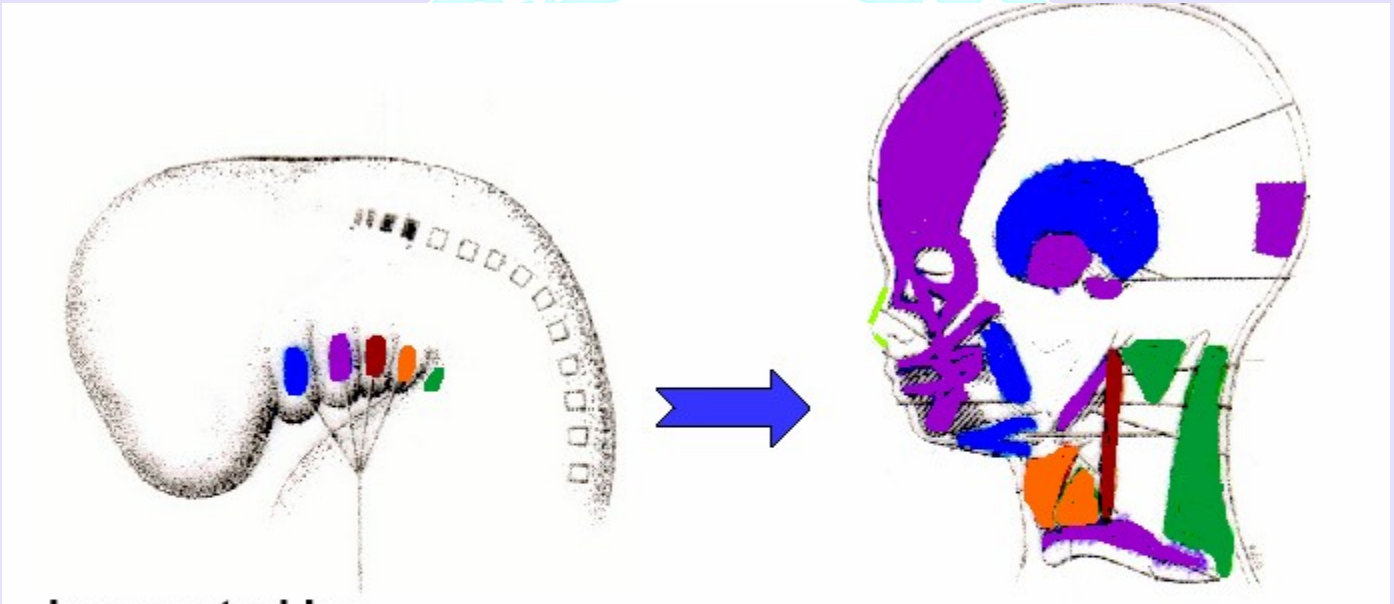
PHARYNGEAL ARCH NERVES.

Muscles of Arches are innervated by Cranial Nerves



1) First Arch – Trigeminal (V)
2) Second Arch – Facial (VII)
3) Third Arch – Glosso-pharyngeal (IX)
4) Fourth Arch – Vagus (X)
5) Caudal Sixth – Accessory (XI)

MUSCLES OF BRANCHIAL ARCHES.



Innervated by

First - Trigeminal V

Second - Facial VII

Third Glosso-pharyngeal IX

Fourth Vagus X

Sixth Accessory XI

SVE = SKELETAL MUSCLES DERIVED FROM BRANCHIAL ARCHES

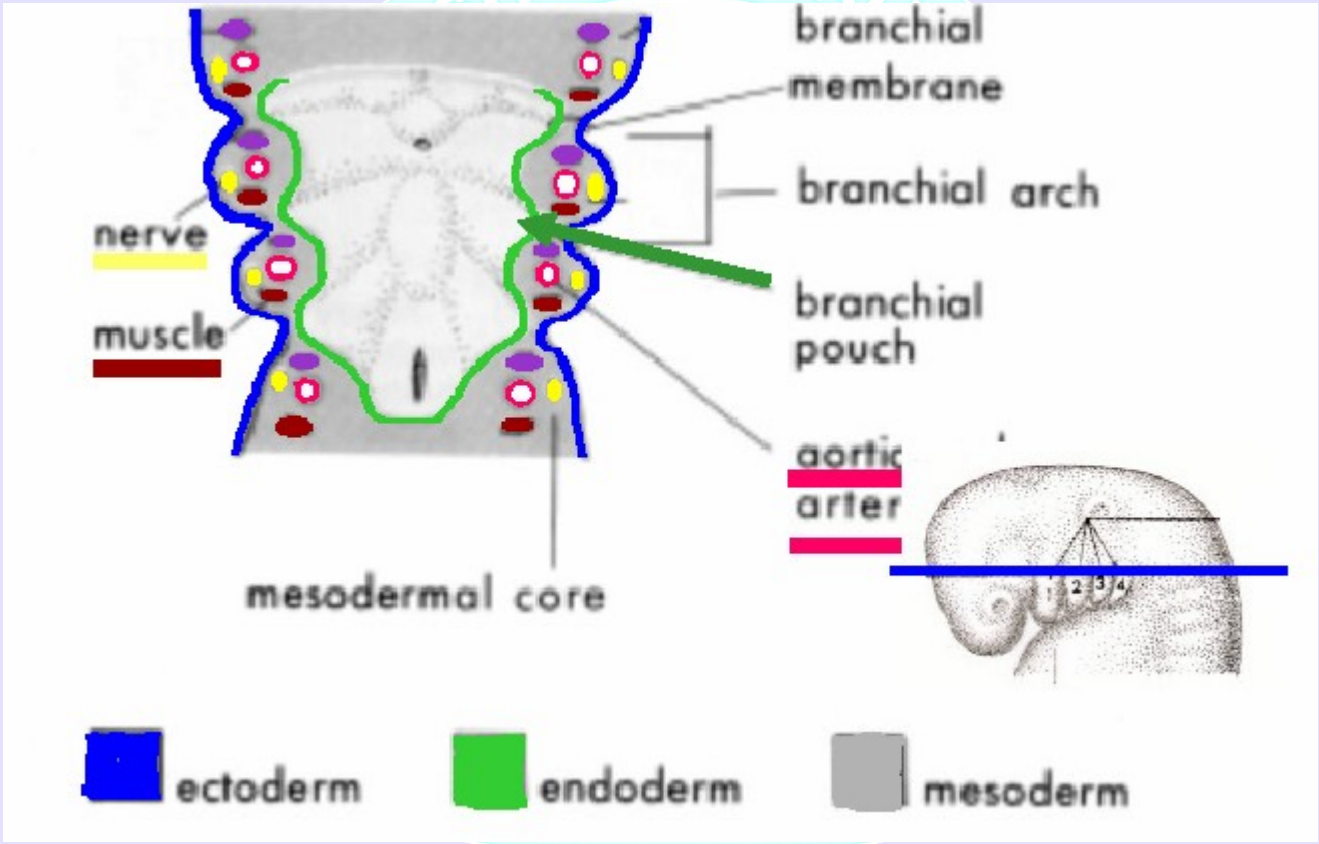
STRUCTURES DERIVED FROM BRANCHIAL ARCHES AND POUCHES

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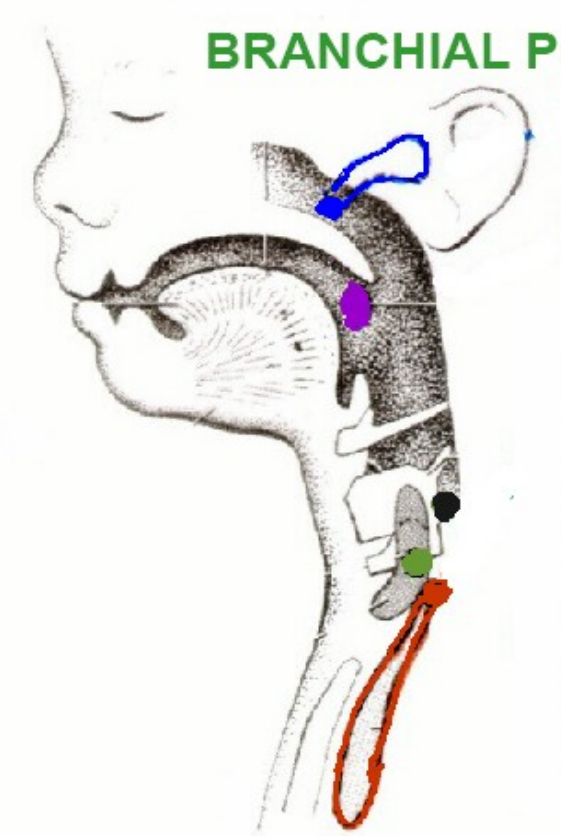
ARCH/NERVE	SKELETAL	LIGAMENTS	MUSCLES	POUCH
First (V)	1) Malleus 2) Incus	1) Ant. ligament of malleus 2) Sphenomandibular ligament	1) Muscles of Mastication 2) Tensor tympani 3) Tensor palati 4) Mylohyoid 5) Ant. belly of Digastric	1) Auditory tube 2) Tympanic cavity
Second (VII)	1) Stapes 2) Styloid process 3) Hyoid bone - lesser horn, upper half of body	Stylohyoid ligament	1) Muscles of Facial Expression 2) Stapedius 3) Stylohyoid 4) Post. belly of Digastric	Lining (crypts) of palatine tonsils
Third (IX)	Hyoid bone - greater horn, lower half of body	------	Stylopharyngeus	1) Inferior parathyroid gland 2) Thymus
Fourth (X)	Cartilages of Larynx	------	1) All muscles of Larynx 2) All muscles of Pharynx (except Stylopharyngeus) 3) All muscles of Soft Palate (except Tensor palati)	1) Superior parathyroid gland 2) C-cells of Thyroid
Sixth (XI)	-------	------	1) Sternocleidomastoid 2) Trapezius	--------

Note: First Branchial Groove (Cleft) becomes External Auditory Meatus
First Branchial Membrane becomes Tympanic Membrane

PHARYNGEAL POUCHES, GROOVES AND MEMBRANES.



BRANCHIAL POUCH DERIVATIVES



A. Pouch 1 – forms
Tubotympanic recess - Auditory Tube, Tympanic cavity

B. Pouch 2 – lining (crypts of Palatine Tonsils

C. Pouch 3- Inferior Parathyroid Glands & thymus

D. Pouch 4 – superior Parathyroid
Glands & C-Cells (Calcitonin)

Note: Pouch 3 derivatives migrate caudal to pouch 4

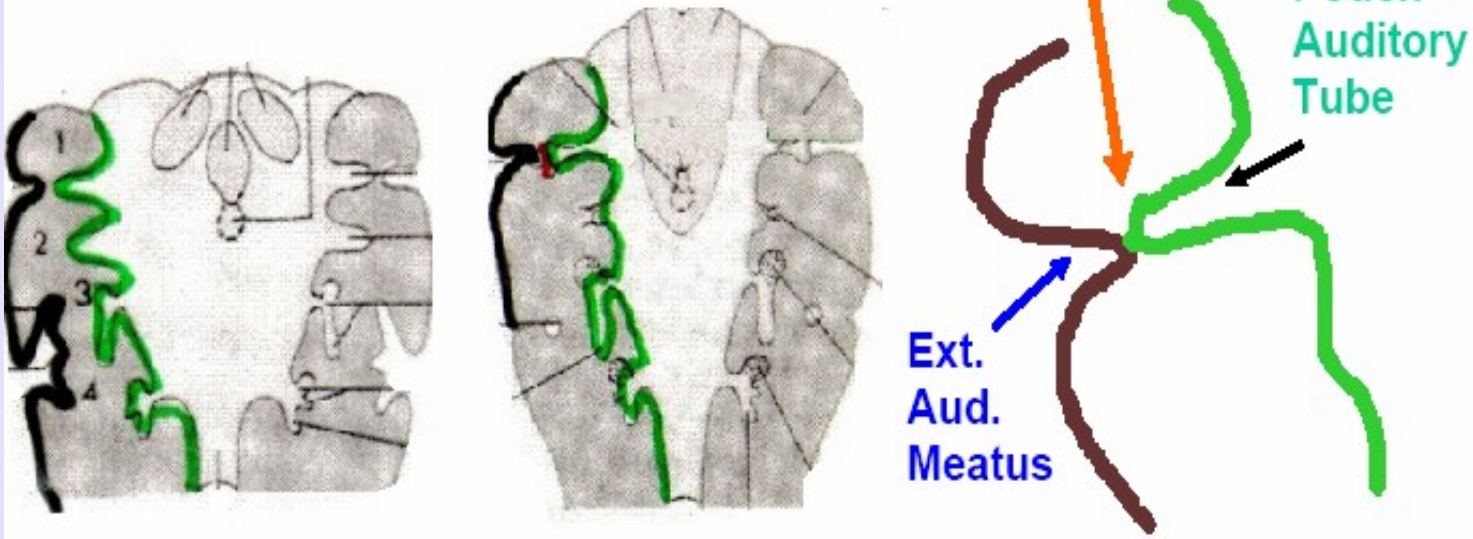
III. PHARYNGEAL GROOVES AND MEMBRANES.

Only First Branchial Groove & Membrane Normally form Structures in Adult

First Groove-External Auditory Meatus

First Membrane — Tympanic Membrane

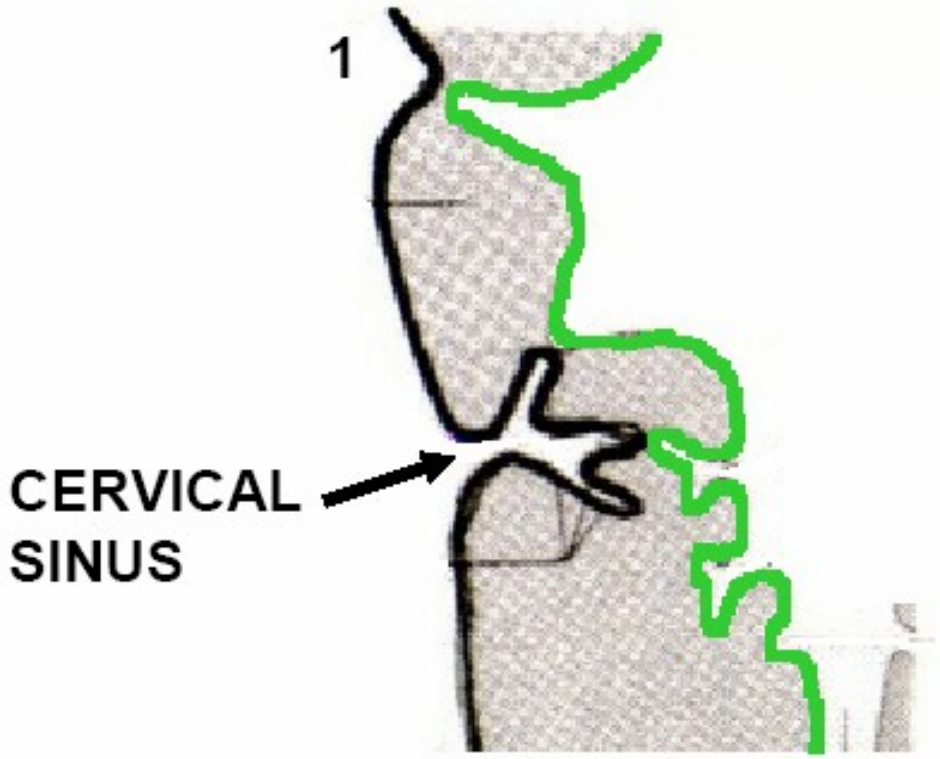
First Pouch - Auditory Tube



PHARYNGEAL GROOVES.

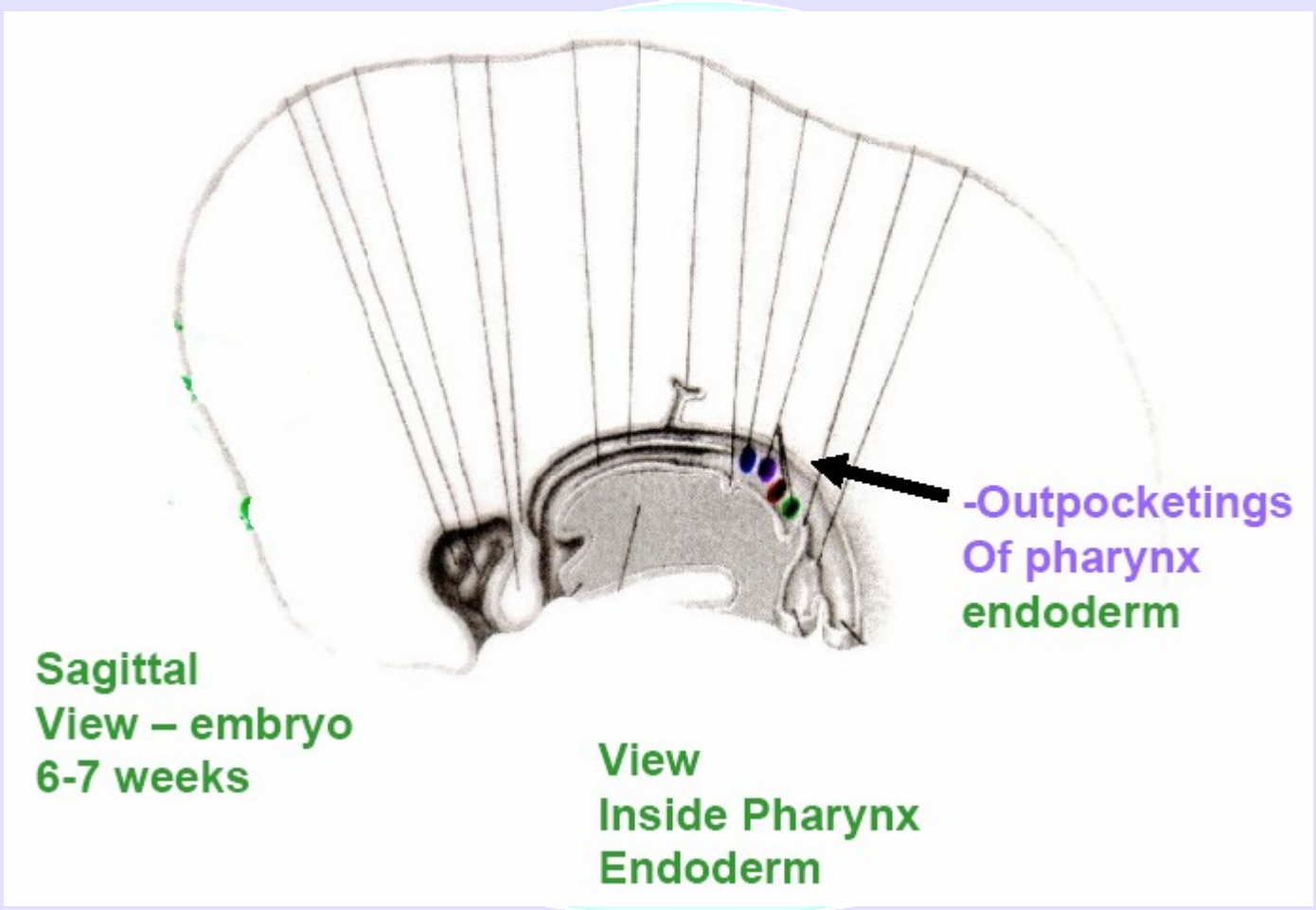
Other Grooves develop in longer depression

Cervical Sinus



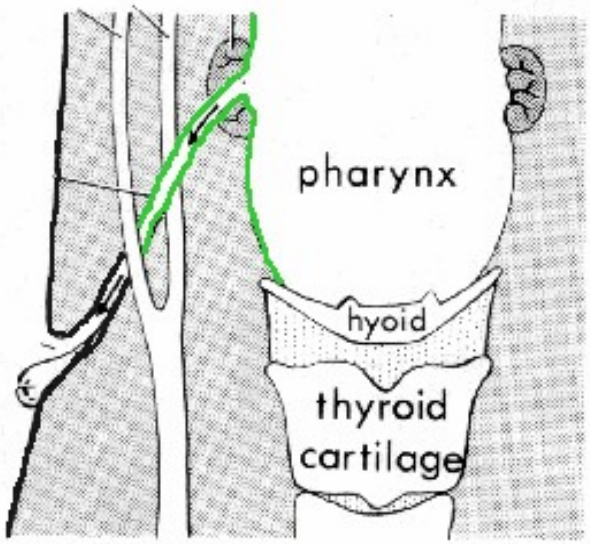
Note:
Cervical sinus normally obliterated but can persist

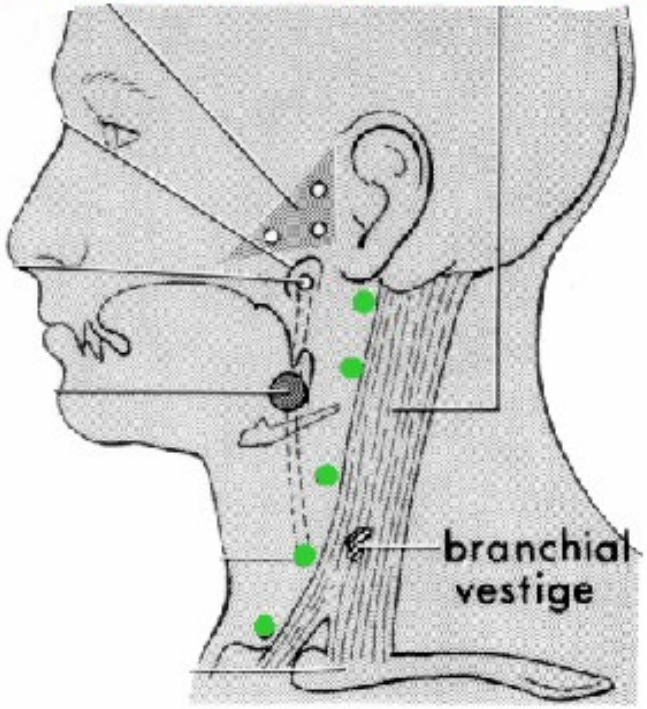
IV. PHARYNGEAL POUCHES.



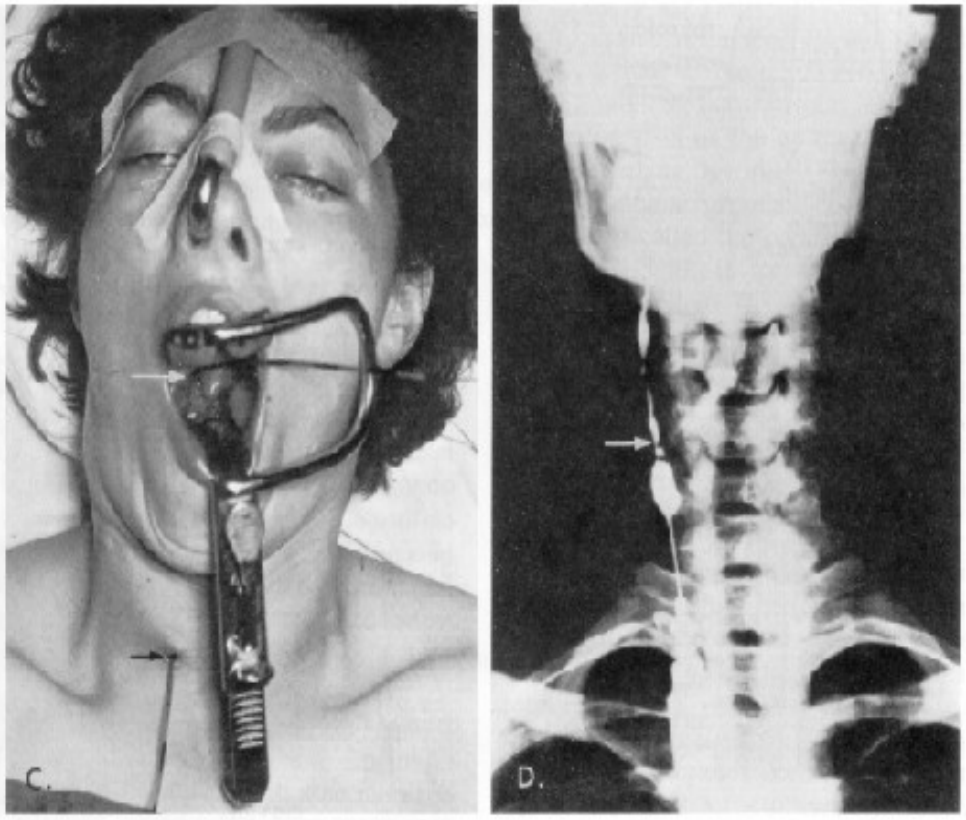
PHARYNGEAL ANOMALIES.

Branchial Sinus = Blind pouch from Pharynx
Branchial Fistula = Channel, often connecting Pharynx to skin of neck; usually passes Anterior to Sternocleidomastoid, between Int. and Ext. Carotid A.

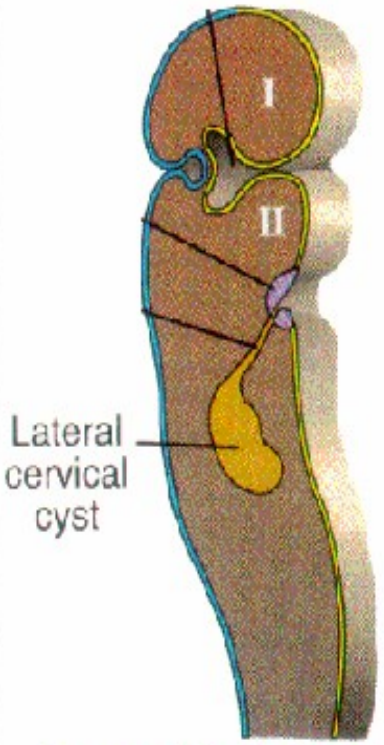




PHARYNGEAL/BRANCHIAL ANOMALIES.



Branchial Fistula - drains to neck



Branchial Cyst often remnant of Cervical Sinus

SELF ASSESSMENT.

1. Define pharyngeal apparatus?
2. What are the different components of pharyngeal apparatus?
3. Each pharyngeal arch is developed from which germ layer?
4. Facial nerve is derived from which pharyngeal arch?
5. Inferior parathyroid gland is developed from which pharyngeal pouch?

THANK YOU ALL.